

Formula Page 2: Metameric Blend Ratio Sweep

Corresponding to Steps 9-12 in calculate_guangpu_metameric_cri.m.

After normalizing the W channel and RGB target white to the same luminance, sweep the blend ratio p:

p = W-channel ratio

1-p = RGB-white ratio

For each p, compute:

- xy

- Y

- CCT

- R1-R8

- Ra

- Rmin8

$$S_{mix} = pS_W + (1 - p)S_{RGB}$$

Final outputs:

$$p_{Ra} = \operatorname{argmax} R_a(p), \quad p_{Rmin} = \operatorname{argmax} R_{min8}(p)$$

- Blend ratio with maximum Ra

- Blend ratio with maximum Rmin8